8.4.2.1 Structure of ATPase (F_1F_0 ATPase)

Mitochondrial ATP synthase is an F-type ATPase. It is quite unique in its structure compared to other ATP synthases derived from chloroplast and eubacteria. Looking into the structure, it is a large enzyme complex found in the inner mitochondrial membrane, which catalyzes the formation of ATP from ADP and Pi. As demonstrated in Fig.8.11, ATP synthase has two discrete components: F_1 (catalytic unit), a peripheral membrane protein, that lies in the mitochondrial matrix and is made up of five type of polypeptide chains (α_3, β_3, γ, ε and δ chains), and F_o (o denoting oligomycin-sensitive), is an integral membrane protein which is composed of three subunits, a, b, and c, in the proportion ab_2c_{10-12}.

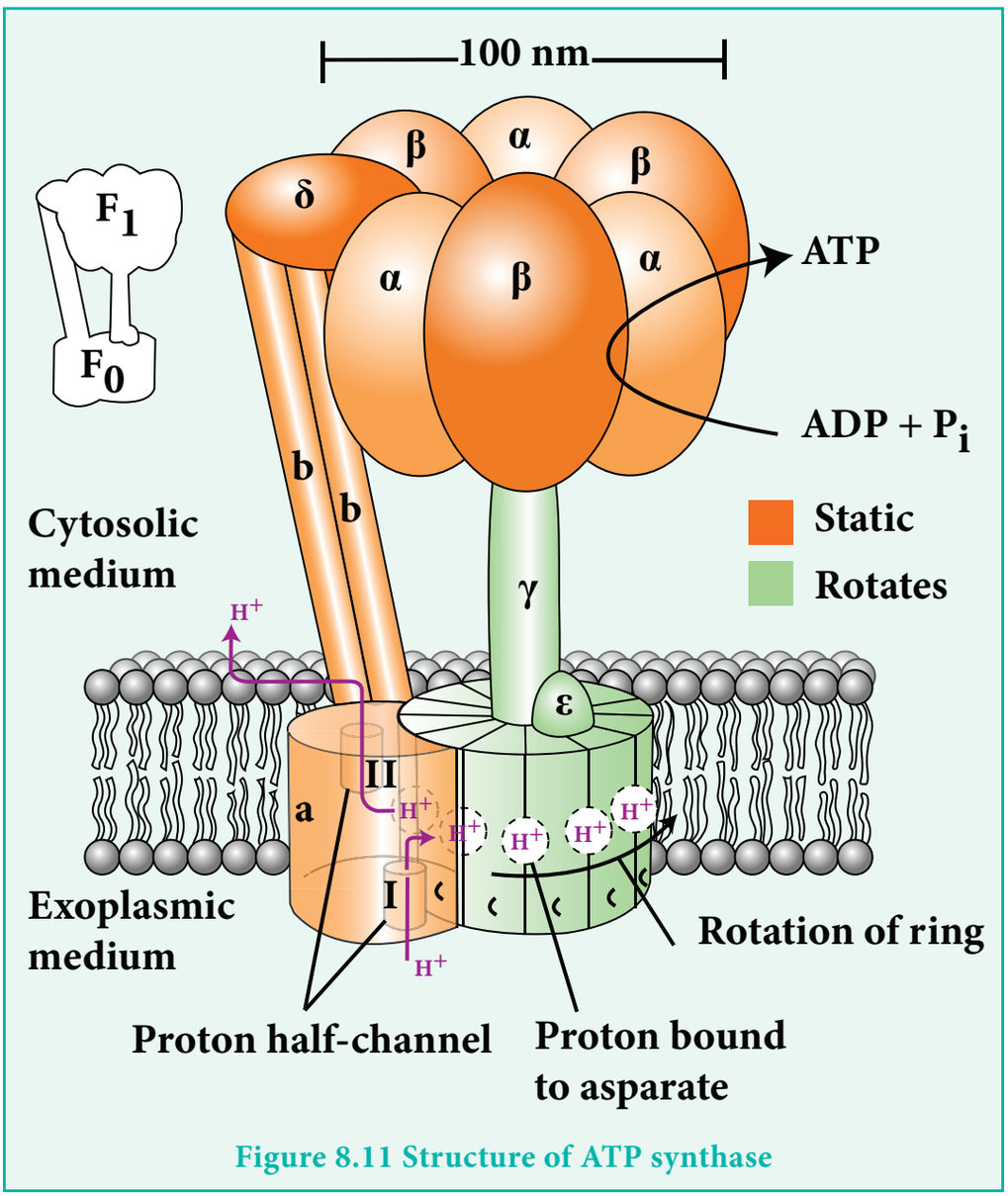


Figure 8.11 Structure of ATP synthase

The critical subunits namely, β possesses ADP / P_i binding sites, α subunit can bind ATP; γ subunit rotates to change to alpha and beta conformations. c subunit of F_0 contains proton half channels; 'a' is the site of proton entry into 'c' subunits.

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